PRICE VISION AI: MULTIMODAL PRODUCT SEARCH AND CROSS-E-COMMERCE REAL-TIME BARGAIN COMPARISON SYSTEM

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ABSTRACT

As e-commerce websites expand rapidly, consumers are confronted with limited ability to find the best deals fragmented prices, concealed shipping fees, uncertain delivery periods, and seller reliability inconsistency. comparison systems mainly rely on the cheapest sticker price, ignoring other factors that affect the ultimate buying paper introduces Price Vision AI, a system powered by AI that allows users to upload an image of a product, identif automatically using vision-language models (CLIP/BLIP), normalize product information, retrieve up-to-date pricin from large online marketplaces (Flipkart, Amazon), and calculate the actual best offer by taking into account final discount, delivery time, shipping fees, and seller reputation. The system is built with FastAPI (backend), Playwrig React (frontend), and Redis/Postgres (data management). The most important features are watchlists, price-drop and historical price trending. Experimental testing shows high recognition accuracy of products (>90%), quick re (<= 6s p95 for two-source comparisons), and stable error handling.

Keywords: *Price Comparison, E-Commerce, Artificial Intelligence, Vision-Language Models, CLIP, BLIP, Playwright, FastAPI.*

I. INTRODUCTION

The international spread of e-commerce sites has dramatically altered the way consumers buy, providing unparallele a staggering diversity of products. Yet the decentralized environment of the contemporary digital economy, control retailers like Flipkart and Amazon, has brought with it an essential challenge: price and deal fragmentation. The typically sort through a confusing array of fluctuating base prices, dynamic coupons, changing shipping charges, a seller reputation to find the actual value proposition of an offer.

State-of-the-art price comparison engines are myopically engineered; they are mostly keyword-based matching- focus exclusively on the lowest sticker price (Pmin) as the top priority, thus overlooking vital drivers of final trans These overlooked factors include variable costs (dynamic shipping fees, time-sensitive coupons) and trust paramet ratings, fulfillment guarantees). Furthermore, traditional text inputs do not align with modern visual search habits. presents Price Vision AI, an AI-based engine engineered to bridge these gaps through multimodal input Multi-Criteria Decision-Making (MCDM).

II. LITERATURE REVIEW

The development of Price Vision AI leverages advancements in three different fields: intelligence (VLM), dynamic web data extraction (browser automation), and decision-making. Conventional price comparison models rely on keyword catalogs and API hidden costs and vendor reliability. To address visual discovery, Contrastive Lang Pre-training (CLIP) maps image vectors to text, and Bootstrapping Language-Image Pre-train yields descriptive product captions to normalize naming variations across platforms. Dynam engines using Playwright solve the limitation of static HTTP requests against JavaScript-rend securely, enabling real-time capture of volatile prices, stock alerts, and promotional coupons.

III. PROPOSED METHODOLOGY & SYSTEM ARCHITECTURE

- **Multimodal Recognition:** Zero-shot product identification utilizing CLIP and BLIP vision-language models to extract high-quality titles from user uploads with 90%+ accuracy.
- **Asynchronous Data Fetching:** Asynchronous Python (asyncio) combined with Playwright browser connectors concurrently scrap Flipkart/Amazon in parallel.
- **Best-Deal Scoring (MCDM):** Implements a weighted Multi-Criteria Decision-Making (MCDM) model, calculating a single final rank score (Si) based on Price, ETA, and Seller Rating using Min-Max scaling.
- **FastAPI & Distributed Queues:** ASGI FastAPI backend serves low-latency requests in parallel. Redis is deployed for scraped data caching and managing concurrent scraping tasks safely.

IV. RESULTS ANALYSIS & EXPERIMENTAL PERFORMANCE

Extensive testing reveals highly competitive performance metrics. Price Vision AI achieved image recognition accuracy of 82.5% and a Top-3 accuracy exceeding 90%. Real-time data across Flipkart and Amazon achieves an end-to-end response delay of <= 6 seconds at percentile, with an average transaction scoring turnaround of 4.2s. Furthermore, the R scraper queue delivers an error recovery success rate above 95% using exponential networ effectively mitigating IP blocks and transient failures.

V. CONCLUSION & REFERENCES

Price Vision AI demonstrates the immense viability of coupling multimodal computer vision with automa scraper networks to empower online shoppers. By scaling from a simple lowest-sticker paradigm to MCDM deal recommendation engine, it provides consumers with a transparent and smart overview deals. Future plans encompass integrating merchant APIs and adding personalized visual alerts.

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